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Q0 a lecture on quantum mechanics

> **The mathematician plays a game in which he himself invents the rules while the physicist plays a game in which the rules are provided by nature, but as time goes on it becomes increasingly evident that the rules which the mathematician finds interesting are the same as those which nature has chosen. -Paul Dirac**

A metaphorical story: I wandered into a bar one evening, took a seat at a table, ordered a beverage, and over the course of the evening found myself an unnoticed audience to a series of conversations

between the regular patrons. The bar, you see, was frequented by physicists speaking a jargon-laden language similar to English. As I had always wanted to understand some physics I listened intently to see what I could make out, but the terminology was dense, layered and far beyond my comprehension. To make matters worse the bar was quite noisy, so much of what was said was obscured by the general din. At the evening's end I found myself sadly puzzled, retaining but a handful of fragmentary ideas, my mind reeling with questions.

Here I offer a *construction*: A framework for quantum mechanics that eventually translates to quantum computing.

From sets to Hilbert space

There are known to be some familiar actors, the subatomic particles: electrons, positrons, neutrons, protons, photons, helium nuclei (α particles), muons; and so on. There are also more exotic particles like neutrinos and whatever dark matter is made from. For these particles to provide us with an entertaining story we need a mathematical stage. So a little ansatz: We must develop *Hilbert space*, returning to the particles only once the mathematical formalism is in place.

Fields

In the mathematical field of *analysis* we begin with a set S, a collection of distinct elements. Add to this two operations *addition* and *multiplication*: To be applied between any two elements of the set. S is *closed* under addition and multiplication if all sums / products are also in S.

Taking such an S to be closed under addition we add four axioms. First: for all a, b, c in S we have $(a+b)+c = a+(b+c)$: **associativity**. Second there is an **identity element** called 0 such that for all a in S we have $a+0=a$. Third we bring in additive inverses: For every a in S there is $-a$ also in S such that $a+(-a)=0$. Fourth addition is **commutative**: For every a and b in S we have $a+b=b+a$. Together with closure all this means that S is an abelian group under addition.

Now we try for the same story for our second operation *multiplication*. Associativity first: $(a\cdot b)\cdot c = a\cdot(b\cdot c)$. Next a multiplicative identity we call "1" (in S). The third axiom hits a snag: We want multiplicative inverses for all a in S but the 0 element we mentioned above, the identity element for addition, has no multiplicative inverse. Why? Because 0 multiplied by anything must be 0. This is a theorem so in the spirit of a thorough narrative let us see this as a proof:

$0\cdot a = (0+0)\cdot a = 0\cdot a + 0\cdot a$, subtract $0\cdot a$ from both sides, get $0 = 0\cdot a$.

A proper inverse of 0 such that $0\cdot 0^{-1} = 1$ is nowhere to be found. So we refine the second 'multiplication' abelian group by stipulating all elements of S stay but for 0. Now for $S\backslash\{0\}$ we have a multiplicative inverse axiom. And fourth axium: Multiplication commutes in abelian groups.

So with a little finagling: We have abelian groups defined for $(S,+)$ and for $(S\backslash\{0\},\times)$. Finally one more axiom is needed that relates addition and multiplication, the distributive axiom: $a(b+c) = a\cdot b + a\cdot c$. (Technically we should state both left and right distributivity, but since multiplication commutes in this field, they are equivalent.) This combination of +-abelian, x-abelian and the distributive axiom brings us to a **field** F.

We have defined a field in this way to provide *scalars*: Elements of a field. This is a prerequisite to defining a *vector space*. In turn a prerequisite to defining a *Hilbert space*. Which in turn will support *states*, vectors used to define quantum systems.

The vector space

A *vector space* V over a field F is a set of elements called 'vectors' that has two operations defined; where these two operations follow certain rules. The first operation is vector addition; so it applies between two elements of V, as in $u + v$. The second operation is 'scalar multiplication' for some scalar $a \in F$. (Technically scalar multiplication is a *map* rather than an *operation*, symbolically $F \times V \to V$.)

The vector space V must satisfy these rules:

- $(V, +)$ is an abelian group: Associativity, Identity, Inverses, Commutativity
- Scalar multiplication $a \cdot v$ follows these four axioms
  - Compatibility with field multiplication: $a(bv) = (ab)v$
  - The field F's multiplicative identity 1 operates trivially: $1 \cdot v = v$
  - Distributivity over vector addition: $a \cdot (u + v) = au + av$
  - Distributivity over field addition: $(a + b)v = av + bv$

So V defined as such is resonating with the definition of the field. The vector space being an abelian group via vector addition echoes the field F. Scalar multiplication for V echoes multiplication in the field F. And Compatibility and Distributivity axioms echo the distributive axiom for F. The shift is in the multiplication by *scalars*, not as an operation between elements of V.

Here is where practical use of vectors (say in linear algebra) often muddles vector space as a mathematical abstraction. The fast track is to introduce dimensionality and vectors as an ordered set of scalars and very quickly we're off to diagrams with arrows and calculations that require scrolling across components. So let's try and stay with the abstraction a bit longer.

A vector $v \in V$ is an abstract entity that obeys the rules of the vector space. There is a thought process by which we produce something called a *basis*, a set of vectors in V that accomplish a task of *representation*. I'll define these below; but the point is that a particular basis is not unique. It can be abandoned in favor of a *different* basis, and so on potentially to *many* other bases. A basis is a subset of V and a formal structure imposed on a vector space that allows us to express any element of V (vectors) by means of representation. This said without reference to dimension or components or $\hat{i}$.

Linear independence, span, basis, dimension

To define "basis" precisely we need two preliminary concepts.

Linear independence: (You can't get there from here principle) A set of vectors $\{v_1, v_2, \ldots, v_k\}$ is linearly independent if the only solution to

$$a_1 v_1 + a_2 v_2 + \cdots + a_k v_k = \mathbf{0}$$

is $a_1 = a_2 = \cdots = a_k = 0$. In other words, no vector in a linearly independent set can be written as a combination of the others.

Span: The span of a set of vectors is the collection of all linear combinations of those vectors. If $\text{span}(\{v_1, \ldots, v_k\}) = V$, the set *spans* V — you can reach every vector in V by combining the k vectors in that set with scalar coefficients. (You can get anywhere principle)

With linear independence and span defined we can now define a basis.

Basis: A set of vectors that is both linearly independent *and* spans V is a *basis* for V. The crucial property: every vector in V can be written *uniquely* as a linear combination of basis vectors and this now includes the 0 vector. Uniqueness (in decomposition) is what makes coordinates (components) well-defined.

Dimension: The number of vectors in any basis of V. A foundational theorem guarantees all bases of a given vector space have the same size, so dimension is well-defined. If V has a finite basis with n elements, $\dim(V) = n$. If no finite basis exists, V is infinite-dimensional.

With a basis $\{e_1, \ldots, e_n\}$ in hand, any vector $v \in V$ can be written as:

$$v = c_1 e_1 + c_2 e_2 + \cdots + c_n e_n$$

where the scalars $c_i \in F$ are the **components** (or coordinates) of v in that basis. Change the basis and the same abstract vector gets different components. The vector is basis-independent; its representation is not.

A vector space can be finite or infinite-dimensional.

Now to this name-dropped but as-yet-undefined space of Hilbert. To the next chapter Q1.

Q1: Hilbert Space

Construction

Q0 built from sets through fields to vector spaces, arriving at the concepts of basis and dimension. Here we add the remaining structure needed to reach a Hilbert space: an inner product, a norm, and completeness.

Definition

A **Hilbert space** $\mathcal{H}$ is a complete inner product space — a vector space equipped with an inner product that is also complete with respect to the norm induced by that inner product.

1. Start with a Vector Space

Begin with a vector space V over the field $\mathbb{F}$ (typically $\mathbb{R}$ or $\mathbb{C}$).

2. Define an Inner Product

An inner product $\langle \cdot, \cdot \rangle : V \times V \to \mathbb{F}$ must satisfy:

- **Conjugate symmetry:** $\langle x, y \rangle = \overline{\langle y, x \rangle}$
- **Linearity in one argument:** $\langle x, \alpha y + \beta z \rangle = \alpha \langle x, y \rangle + \beta \langle x, z \rangle$
- **Positive definiteness:** $\langle x, x \rangle \geq 0$ with equality iff $x = 0$

(Convention note: physics places linearity in the *second* argument as shown here; some math texts place it in the first. The distinction matters for bookkeeping but not for the structure of the theory.)

3. Induce a Norm

The inner product induces a norm:

$$\|x\| = \sqrt{\langle x, x \rangle}$$

Note here that the inner product produces a scalar; so now we are back in the supporting Field. But the norm (built on a special case of the inner product) produces a scalar from $\mathbb{R}$ that is greater than or equal to 0. (That's the positive definiteness in play.) The square root by convention refers to the positive root.

This norm satisfies the **parallelogram law**:

$$\|x + y\|^2 + \|x - y\|^2 = 2\|x\|^2 + 2\|y\|^2$$

4. Complete the Space

A space is **complete** if every Cauchy sequence converges within the space. If the inner product space is not already complete, we take its completion:

- Identify all Cauchy sequences in V
- Define equivalence classes of Cauchy sequences (two sequences are equivalent if their difference converges to zero)
- The set of equivalence classes, with the extended inner product, forms the Hilbert space $\mathcal{H}$

Example: an inner product space that is not complete

Let V be the set of all continuous functions $f : [0, 1] \to \mathbb{R}$, with inner product:

$$\langle f, g \rangle = \int_0^1 f(x)\, g(x)\, dx$$

This satisfies all the inner product axioms (linearity, symmetry, positive-definiteness). It is a perfectly valid inner product space — but it is **not** complete.

Consider the sequence of continuous functions:

$$f_n(x) = \begin{cases} 0 & \text{if } x \in [0, \frac{1}{2} - \frac{1}{n}] \\ \text{linear ramp} & \text{if } x \in [\frac{1}{2} - \frac{1}{n}, \frac{1}{2} + \frac{1}{n}] \\ 1 & \text{if } x \in [\frac{1}{2} + \frac{1}{n}, 1] \end{cases}$$

Each f_n is continuous (the ramp connects 0 to 1 smoothly). As $n \to \infty$, the ramp gets steeper and the functions "try to converge" to the step function:

$$f(x) = \begin{cases} 0 & x < \frac{1}{2} \\ 1 & x > \frac{1}{2} \end{cases}$$

This is a Cauchy sequence in the L^2 norm ($\|f_n - f_m\| \to 0$ as $n, m \to \infty$), but the limit f is **not continuous** — it has a jump at $x = 1/2$. The limit does not live in V. The sequence converges to something *outside the space*. There is a "hole."

The fix: enlarge the space to $L^2([0,1])$ — the space of all square-integrable functions (not just continuous ones). Now step functions are included, the Cauchy sequence has its limit, and the space is complete. $L^2([0,1])$ with this inner product is a Hilbert space.

The analogy is: continuous functions on [0,1] are to L^2 as $\mathbb{Q}$ is to $\mathbb{R}$. Completion fills the holes.

Examples

Space	Inner Product	Notes
$\mathbb{R}^n$	$\langle x, y\rangle = \sum_{i=1}^{n} x_i y_i$	Finite-dimensional, automatically complete
ℓ^2	$\langle x, y\rangle = \sum_{i=1}^{\infty} x_i \overline{y_i}$	Space of square-summable sequences
$L^2([a,b])$	$\langle f, g\rangle = \int_a^b f(t)\overline{g(t)}\, dt$	Completion of continuous functions

Important Properties

1. **Riesz Representation Theorem:** Every bounded linear functional on $\mathcal{H}$ can be represented as an inner product with a fixed element.
2. **Orthonormal Bases:** Every Hilbert space admits an orthonormal basis $\{e_i\}$ such that any $x \in \mathcal{H}$ can be written as:
$$x = \sum_i \langle x, e_i\rangle\, e_i$$
3. **Projection Theorem:** For any closed subspace $M \subseteq \mathcal{H}$, every element x has a unique decomposition:
$$x = x_M + x_{M^\perp}, \quad x_M \in M, \quad x_{M^\perp} \in M^\perp$$

Why Completeness Matters

Without completeness, we lose: - Convergence guarantees for orthogonal expansions - The projection theorem (closest point in a closed subspace) - The spectral theorem for self-adjoint operators - The foundation for quantum mechanics, where states live in $\mathcal{H}$

Connection to Quantum Mechanics: states as vectors

This is the foundational postulate: every *physical state* of a quantum system corresponds to a **ray** in a Hilbert space (a ray being an equivalence class of vectors differing only by a nonzero scalar multiple). After normalizing and factoring out global phase, we say "a state *is* a unit vector in Hilbert space."

For a single qubit: the Hilbert space is $\mathbb{C}^2$ (two-dimensional, complex). A state is a unit vector $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ with $|\alpha|^2 + |\beta|^2 = 1$.

For a system with more degrees of freedom (e.g. a particle moving in space), the Hilbert space is infinite-dimensional ($L^2(\mathbb{R})$, the space of square-integrable functions), but the principle is the same: one state = one (ray of) vector(s).

Dirac notation and the inner product

The inner product defined above in mathematician's notation ($\langle a, b\rangle$) is the same object as Dirac's "bra-ket" notation ($\langle a|b\rangle$), standard in physics.

Both denote a map that takes two vectors and produces a complex number, satisfying:

1. **Conjugate symmetry:** $\langle a|b\rangle = \overline{\langle b|a\rangle}$
2. **Linearity in the second argument:** $\langle a|(\alpha b + \beta c)\rangle = \alpha\langle a|b\rangle + \beta\langle a|c\rangle$
3. **Positive definiteness:** $\langle a|a\rangle > 0$ for $|a\rangle \neq 0$

(Note: physics convention puts linearity in the *second* slot and conjugate-linearity in the first. Some math texts reverse this — the object is the same.)

For the qubit case with $|a\rangle = (a_0, a_1)^T$ and $|b\rangle = (b_0, b_1)^T$:

$$\langle a|b\rangle = \bar{a}_0 b_0 + \bar{a}_1 b_1$$

where $\bar{a}_0$ denotes the complex conjugate. This is the standard Hermitian inner product on $\mathbb{C}^2$.

Dirac's notation splits a vector into a "ket" $|b\rangle$ (column vector) and a "bra" $\langle a|$ (conjugate-transpose row vector), so that the inner product is literally matrix multiplication: row times column gives a scalar.

Q2: The Hadamard gate as 2-point DFT

Q0 built the abstract mathematics (field → vector space → basis). Q1 completed the Hilbert space (inner product → norm → completeness) and connected it to quantum mechanics via the foundational postulate (state = unit vector) and Dirac notation. Here in Q2 we work concretely with the simplest quantum system — a single qubit — developing the operational vocabulary of quantum computing: gates, measurement, the Bloch sphere, and the relationship between the Hadamard gate and the discrete Fourier transform.

Spadework

A certain amount of terminology is introduced in what follows without a great deal of explanation. This is in the spirit of 'time exists so that everything doesn't happen at once': The intent is to elaborate upon and fit together these terms and ideas in the subsequent narrative, ultimately arriving at a coherent framework.

This document covers, in order:

1. **Spadework** — terminology, workflow, Pauli matrices, unitary/Hermitian definitions
2. **DFT and the Hadamard** — the 2-point DFT as a basis change
3. **Bases and eigenstates** — Z-basis, X-basis, complex amplitudes, $\mathbb{R}^2$ vs $\mathbb{C}^2$
4. **The Bloch sphere** — visualizing states, phase decomposition, global phase degeneracy
5. **Multi-qubit systems** — tensor products, entanglement

6. **Gates in action** — active/passive pictures, Pauli rotations, dual role as operators

To cut to the chase, then: Most quantum computing algorithms follow a standard workflow: An execution plan of 'initialize - transform - measure - repeat' - Initialization means establishing a quantum system - This is in the form of some number of quantum bits (qubits) - Transformation is the application of a series of discrete logical operations call *gates* - The application of these gates effects a *time evolution* of the quantum system - Measurement resolves the qubits into classical binary states - Repetition accumulates measurements many times to arrive at a probability distribution

There are refinements of and exceptions to this strategy; but for now the narrative will build towards this standard workflow, starting with some spadework of the 'terms defined without a great deal of explanation' variety.

States, operators and measurement

We imagine a quantum system as a simple construction – starting with a single qubit. The quantum system has a *state* that in practice will be a vector in a vector space (in fact a particular Hilbert space). This state will be mathematically abstract. There are two types of operators that are regularly applied to states: Gates and observables. Gates are necessarily unitary operators and observables are necessarily Hermitian operators. Both are applied to quantum states in the course of carrying out a quantum calculation.

This said: There is a separate notion of *measurement* as a *procedure* that invokes operators but also introduces probability and irreversibility.

The Pauli spin matrices

In quantum computing we work with qubits understood to have two measureable states. We assert without proof that a number of 2 x 2 matrices will play an important role. Here we introduce three 2x2 unitary Hermitian matrices – the Pauli spin matrices – in terms of components

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Each has eigenvalues $+1$ and -1. Their eigenvectors define three mutually complementary bases for a single qubit (the X, Y, and Z bases). They generate rotations on the Bloch sphere (described below) about the x, y, and z axes respectively.

Unitary and Hermitian matrices: definitions

Hermitian (self-adjoint): A matrix A is Hermitian if it equals its own conjugate transpose: $A = A^\dagger$, where $A^\dagger$ is obtained by transposing A and taking the complex conjugate of every entry.

- All eigenvalues are **real**
  - In QM observables must be Hermitian — measurement outcomes are real numbers
- Eigenvectors corresponding to distinct eigenvalues are mutually orthogonal
- This is easily verified by inspection for the Pauli matrices

Unitary: A matrix U is unitary if its conjugate transpose is its inverse: $U^\dagger U = UU^\dagger = I$.

In 3B1B Linear Algebra this relates to 'what happens to the unit cell upon transform?'

- U **preserves norms**: $\|U|\psi\rangle\| = \||\psi\rangle\|$.
  - A unit vector stays a unit vector.
  - This is why gates must be unitary — they preserve the normalization (total probability = 1).
- U **preserves inner products**: $\langle U\phi|U\psi\rangle = \langle\phi|\psi\rangle$.
  - Angles and orthogonality are unchanged.
- Geometrically: unitary = rotation (possibly with reflection) of the Hilbert space.
  - No stretching or squishing.

Example: $\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. Compute $\sigma_x^\dagger \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$.

The connection: Unitary operators are the allowed *time evolutions* and *gates* (they preserve probability). Hermitian operators are the allowed *observables* (they have real measurement outcomes). The Pauli matrices happen to satisfy both conditions simultaneously, which is unusual and useful.

Discrete Fourier Transform (DFT)

The DFT operates on an n-dimensional vector producing a new n-vector. Each element of this new vector is a weighted sum of the elements of the original vector. Hence we need two indices: k as the index of an element of the output vector and j when indexing the elements of the input vector.

With symmetric $1/\sqrt{2}$ normalization and $n = 2$:

$$X_k = \frac{1}{\sqrt{2}} \sum_{j=0}^{1} x_j \, e^{-\pi i \, jk}$$

For $k = 0$: the phasor is $e^0 = 1$ for both $j = 0$ and $j = 1$. So $X_0 = \frac{1}{\sqrt{2}}(x_0 + x_1)$.

For $k = 1$: the phasor is $e^0 = 1$ for $j = 0$, and $e^{-\pi i} = -1$ for $j = 1$. So $X_1 = \frac{1}{\sqrt{2}}(x_0 - x_1)$.

Written as a matrix:

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

In quantum mechanics this is the Hadamard gate: The 2-point DFT with symmetric normalization.

What is the basis in this situation?

There are two bases in play, and the DFT/Hadamard is the map between them:

The "time domain" basis is the standard/computational basis: $\{|0\rangle, |1\rangle\}$ — the vectors $(1, 0)^T$ and $(0, 1)^T$. When we write a state as $\alpha|0\rangle + \beta|1\rangle$, the components α and β are "time-domain samples." In quantum computing this is the Z-basis — the eigenbasis of σ_z.

The "frequency domain" basis is $\{|+\rangle, |-\rangle\}$ — the vectors $\frac{1}{\sqrt{2}}(1, 1)^T$ and $\frac{1}{\sqrt{2}}(1, -1)^T$. These are the rows of the Hadamard matrix (equivalently, the DFT basis vectors — the phasors themselves). In quantum computing this is the X-basis — the eigenbasis of σ_x.

The key insight: the DFT basis vectors (the phasors you sum against) *become* the new basis. When you compute the DFT, you project your vector onto each phasor — and those projections are the components in the frequency basis. The Hadamard/DFT doesn't create a new vector; it re-expresses the *same* vector in terms of these phasor-basis states.

So the "time domain" / "frequency domain" distinction is just a change of basis in $\mathbb{C}^2$. The Hadamard is the unitary matrix that implements this change. This is the same structure as position/momentum in continuous QM, shrunk down to two dimensions.

The Z-basis as the eigenbasis of σ_z

Recall σ_z from the Spadework section. Its eigenvectors satisfy $\sigma_z|v\rangle = \lambda|v\rangle$:

- $\sigma_z|0\rangle = (+1)|0\rangle$ — eigenvalue $+1$
- $\sigma_z|1\rangle = (-1)|1\rangle$ — eigenvalue -1

So "the Z-basis is the eigenbasis of σ_z" means: $\{|0\rangle, |1\rangle\}$ are the vectors that σ_z leaves unchanged in direction (scaled by their eigenvalue).

Structurally $|0\rangle = (1,0)^T$ and $|1\rangle = (0,1)^T$ are the standard unit vectors, analogous to $\hat{i}$ and $\hat{j}$ in $\mathbb{R}^2$. The crucial difference: here they span $\mathbb{C}^2$, not $\mathbb{R}^2$. The scalars are complex numbers.

Complex amplitudes

A general single-qubit state is:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad \alpha, \beta \in \mathbb{C}, \quad |\alpha|^2 + |\beta|^2 = 1$$

The complex nature of α and β is essential — not a generalization but a requirement. States with the same *magnitudes* of α and β but different *relative phases* are physically distinct:

- $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ — relative phase 0
- $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$ — relative phase π
- $|+i\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle)$ — relative phase $\pi/2$

These are three different points on the Bloch sphere equator: X, −X, and Y directions respectively. Without complex scalars, the Y-axis vanishes and you cannot represent the full state space.

From $\mathbb{R}^2$ to $\mathbb{C}^2$: Why the scalar field matters

Consider σ_z first in $\mathbb{R}^2$. It acts as reflection across the x-axis: $(x, y)^T \to (x, -y)^T$. The eigenvectors are directions unchanged (up to sign) by the transform: $\hat{i} = (1,0)^T$ lies along the mirror (eigenvalue $+1$) and $\hat{j} = (0,1)^T$ is perpendicular to it (eigenvalue -1, i.e. flipped).

Now transition to $\mathbb{C}^2$. The basis vectors $(1,0)^T$ and $(0,1)^T$ look identical. The eigenvector/eigenvalue structure of σ_z is unchanged. What changes is the **scalar field** supplying the coefficients:

	$\mathbb{R}^2$	$\mathbb{C}^2$
Scalar field	$\mathbb{R}$	$\mathbb{C}$
Basis vectors	$(1,0)^T$, $(0,1)^T$	$(1,0)^T$, $(0,1)^T$

	$\mathbb{R}^2$	$\mathbb{C}^2$
General vector	$a\hat{i} + b\hat{j}$, two real params	$\alpha\|0\rangle + \beta\|1\rangle$, four real params
σ_z eigenvalues	$+1$, -1	$+1$, -1
Non-eigenvector states	(a, b) with both nonzero	(α, β) with complex phases — richer

The basis vectors *themselves* don't change — what changes is the set of states they can span. With complex scalars you can form $\frac{1}{\sqrt{2}}(|0\rangle + e^{i\phi}|1\rangle)$ for arbitrary phase ϕ. These states have no analog in $\mathbb{R}^2$ because $e^{i\phi}$ is not a real number.

The physics demands $\mathbb{C}$ because interference requires phase. Two amplitudes equal in magnitude but differing in phase can cancel (destructive interference) or reinforce (constructive interference). This continuum of partial interference is experimentally observed and cannot be reproduced with real scalars alone.

Visualizing global phase degeneracy

From , to , , — slowly

Starting from the general single-qubit state established above ($|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ with $|\alpha|^2 + |\beta|^2 = 1$), we now decompose and to reveal the Bloch sphere parameters.

Since and are complex, write them in polar form:

$$\alpha = r_\alpha\, e^{i\eta}, \quad \beta = r_\beta\, e^{i\nu}$$

where $r_\alpha, r_\beta \geq 0$ are magnitudes and η, ν are phase angles. Normalization says $r_\alpha^2 + r_\beta^2 = 1$.

Now factor out the phase of from the entire state:

$$|\psi\rangle = r_\alpha e^{i\eta}|0\rangle + r_\beta e^{i\nu}|1\rangle = e^{i\eta}\left(r_\alpha|0\rangle + r_\beta e^{i(\nu-\eta)}|1\rangle\right)$$

Define:

- = — this is the global phase (the phase we factored out)
- = − — this is the relative phase (the difference between the two component phases)
- Since $r_\alpha^2 + r_\beta^2 = 1$, parameterize: $r_\alpha = \cos\frac{\theta}{2}$, $r_\beta = \sin\frac{\theta}{2}$ for some $\theta \in [0, \pi]$

Result:

$$|\psi\rangle = e^{i\gamma}\left(\cos\frac{\theta}{2}|0\rangle + e^{i\phi}\sin\frac{\theta}{2}|1\rangle\right)$$

So to summarize the translation:

Original	Bloch colatitude	Bloch azimuth	Not appearing (global phase)
r_α (magnitude of)	$r_\alpha = \cos\frac{\theta}{2}$	—	—

Original	Bloch colatitude	Bloch azimuth	Not appearing (global phase)
r_β (magnitude of)	$r_\beta = \sin\frac{\theta}{2}$	—	—
(phase angle of)	—	—	= (unobservable)
(phase angle of)	—	= −	(absorbed into and)

The magnitudes of and jointly determine — how far the state is from the north pole. The *difference* of their phase angles gives — the azimuthal position. The *absolute* phase angle of becomes — invisible on the Bloch sphere.

Two complex planes: visualizing and

You can visualize the state as a point in *two* complex planes simultaneously:

Plane 1 ( ): Draw a unit circle. Place somewhere on or inside it at radius r_α and angle . This fixes the "how much $|0$ " and "what phase for the $|0$ component."

Plane 2 ( ): Draw another unit circle. Place at radius r_β and angle . This fixes the "how much $|1$ " and "what phase for the $|1$ component."

The constraint $r_\alpha^2 + r_\beta^2 = 1$ means: if you fix r_α, then $r_\beta = \sqrt{1 - r_\alpha^2}$. So 's radius is determined by 's radius. But their angles and are independent — giving two free angular parameters.

Now the key decomposition:

- (polar angle on Bloch sphere) is determined by r_α alone: $\theta = 2\arccos(r_\alpha)$. Since $r_\beta = \sqrt{1 - r_\alpha^2}$, knowing one radius fixes the other — so equivalently encodes the *balance* between the two components. North pole: $r_\alpha = 1, r_\beta = 0$. South pole: $r_\alpha = 0, r_\beta = 1$. Equator: $r_\alpha = r_\beta = 1/\sqrt{2}$.
- (azimuthal angle on Bloch sphere) is the *difference* of the two plane angles: $\phi = \nu - \eta$. This is physically meaningful — it's measurable.
- (global phase, unobservable) is the *absolute* angle . You can rotate both points in their respective planes by the same angle and nothing physical changes.

Why is azimuthal when the basis vectors are along z?

This is a reasonable confusion. The basis vectors $|0$ and $|1$ sit at the north and south poles — separated by polar angle along the z-axis. There's no azimuthal content in the basis vectors themselves. So where does the azimuthal angle come from?

The answer: is not an angle *between* the basis vectors. It's the *relative phase* between the *coefficients* in the superposition. When both coefficients are nonzero (i.e. the state is a superposition, not a pure basis state), the relative phase determines *which* superposition you have among the continuum of equal-weight superpositions.

Geometrically on the Bloch sphere:

- At the poles (pure $|0$ or pure $|1$), one coefficient is zero so is undefined — there's no azimuthal position. This is like the longitude being undefined at Earth's geographic poles.

- On the equator ($\theta = \pi/2$, i.e. $r_\alpha = r_\beta = 1/\sqrt{2}$), sweeps you through notable states that form two additional bases:
  - $|+\rangle$ ($= 0$) and $|-\rangle$ ($=$) — these are the **X-basis** (eigenstates of σ_x)
  - $|+i\rangle$ ($=$ $/2$) and $|-i\rangle$ ($= 3$ $/2$) — these are the **Y-basis** (eigenstates of σ_y)

  Together with the Z-basis $\{|0\rangle, |1\rangle\}$ at the poles, these give three mutually complementary bases — one for each Pauli matrix, one for each axis of the Bloch sphere.

At intermediate latitudes, still sweeps a circle but the states are no longer equal-weight superpositions.

So parameterizes *which direction the superposition points* in the xy-plane of the Bloch sphere. It's a property of the superposition coefficients, not of the basis vectors.

Is global phase visible on the Bloch sphere?

No — and that's the point. The Bloch sphere has already had global phase removed. Every point on the Bloch sphere represents an *equivalence class* of states that differ only by global phase. The sphere is a 2D surface (parameterized by and) precisely because the third parameter ( ) has been declared unphysical and projected away.

If you want to visualize where "went": it's the invisible extra dimension at each point of the sphere. The full state space is 3-dimensional (as a real manifold): , , and . The Bloch sphere is the 2D *shadow* you get when you ignore .

An equivalent picture: the 3-sphere S^3 in four dimensions

Everything above was developed using two complex planes with a normalization constraint. There is an equivalent, more abstract way to say the same thing: the set of all normalized single-qubit states forms S^3, the **3-sphere**, which lives in **four-dimensional** space.

To be precise about what this means — and to avoid confusion with an ordinary ball in $\mathbb{R}^3$:

- S^1 (a circle) is a 1D curve embedded in $\mathbb{R}^2$. It is the set of points at unit distance from the origin in 2D.
- S^2 (an ordinary sphere, like a basketball) is a 2D surface embedded in $\mathbb{R}^3$. It is the set of points at unit distance from the origin in 3D.
- S^3 is a **3D hypersurface** embedded in $\mathbb{R}^4$. It is the set of points at unit distance from the origin in 4D.

S^3 cannot be visualized as a single object in 3D space. It is not a solid ball, not a sphere you can hold — it is inherently four-dimensional.

Why it appears here: A single-qubit state $(\alpha, \beta) \in \mathbb{C}^2$ has four real components (since $\mathbb{C}^2 \cong \mathbb{R}^4$). The normalization constraint $|\alpha|^2 + |\beta|^2 = 1$ restricts the state to the set of unit vectors in $\mathbb{R}^4$ — which is exactly the definition of S^3.

Connection to the two-planes picture: the two complex planes (one for , one for) with the constraint $r_\alpha^2 + r_\beta^2 = 1$ are a concrete coordinatization of S^3. They are two different ways of saying the same thing: one geometric (a hypersurface in 4D), one computational (two linked complex planes).

From S^3 to the Bloch sphere: the global phase degeneracy — rotating both and by the same angle ($\to$ $+$, $\to$ $+$) — leaves the physics unchanged because $=$ $-$ is invariant.

That synchronized rotation traces a circle within S^3. Collapsing each such circle to a single point reduces S^3 (3-dimensional) to S^2 (2-dimensional) — the Bloch sphere. In the language of topology this is the **Hopf fibration**: $S^3 \to S^2$ with circular fibers, and the Bloch sphere is the base space.

Summary of the logical chain

$$\alpha, \beta \in \mathbb{C} \xrightarrow{\text{polar form}} r_\alpha e^{i\eta},\ r_\beta e^{i\nu} \xrightarrow{\text{normalize}} r_\alpha^2 + r_\beta^2 = 1$$

$$\xrightarrow{\text{parameterize radii}} \theta \xrightarrow{\text{take phase difference}} \phi = \nu - \eta \xrightarrow{\text{discard absolute phase}} \gamma = \eta \text{ (unobservable)}$$

$$\xrightarrow{\text{Bloch sphere}} (\theta, \phi) \text{ specifies the physical state}$$

Convention: global phase of eigenstates

When we write the eigenstates of σ_x as:

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle), \quad |-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

the global phase has been set to zero **by convention** — meaning is chosen to be real and positive ($\eta = 0$). This is not physics but a choice of representative from the equivalence class of states that differ only by global phase.

For $|-\rangle$ specifically: $\alpha = 1/\sqrt{2}$ (real, positive) and $\beta = -1/\sqrt{2} = \frac{1}{\sqrt{2}}e^{i\pi}$. So $r_\beta = 1/\sqrt{2}$ and $\nu = \pi$. The relative phase is $\phi = \nu - \eta = \pi - 0 = \pi$, placing it at the −x point on the Bloch sphere.

The general principle: eigenvectors are only determined up to a scalar multiple by the eigenvalue equation $A|v\rangle = \lambda|v\rangle$. We conventionally normalize (magnitude 1) and fix the global phase (first nonzero component real and positive). This selects one representative from the degenerate circle of physically identical states.

Invariance, symmetry, and conservation

Noether's theorem: every continuous symmetry of a physical system corresponds to a conserved quantity. How does this connect to the phase structure of a qubit?

Global phase symmetry → probability conservation:

- **Symmetry:** physics is invariant under $|\psi\rangle \to e^{i\gamma}|\psi\rangle$
- **Conserved quantity:** total probability ($|\alpha|^2 + |\beta|^2 = 1$ is preserved under all unitary time evolution). In quantum field theory, this same U(1) symmetry gives conservation of electric charge.

This is a genuine Noether-type connection.

Relative phase is not a symmetry:

The relative phase = − is a *dynamical variable*, not a symmetry. Different values of correspond to different physical states with different measurement outcomes. can change under time evolution (any Hamiltonian with a σ_z component will rotate).

is gauge-invariant but not conserved:

The statement “ is invariant under global phase shifts” means: is a well-defined physical observable that does not depend on the arbitrary choice of convention. This is a statement about the *definition* of being robust (gauge-invariant), not a conservation law.

	Symmetry/invariance	Associated conservation
Global phase	Physics unchanged under $\|\psi\rangle \to e^{i\gamma}\|\psi\rangle$	Total probability / charge
Relative phase	Not a symmetry — different = different physics	Not conserved; evolves in time
invariant under -shifts	is gauge-invariant (well-defined physical quantity)	Not a conservation law

The qubit as the atomic unit of quantum computing

Is the qubit the only fundamental building block?

Essentially yes. In the current landscape of quantum computing, the qubit (a two-level quantum system, Hilbert space $\mathbb{C}^2$) is the universal atomic unit. All major platforms — IBM, Google, IonQ, Quantinuum, Rigetti, QuEra — are built on qubits.

There is research into **qutrits** ($\mathbb{C}^3$, three-level systems) and more generally **qudits** ($\mathbb{C}^d$, d-level systems), but these are not commercially available and remain a niche research direction. For practical purposes: qubit = atom of quantum computation.

Multi-qubit systems build on the single-qubit foundation

Yes — and the construction is precise. Once you understand a single qubit (state in $\mathbb{C}^2$, Bloch sphere, gates as rotations), multi-qubit systems are built via the **tensor product**:

- 1 qubit: Hilbert space is $\mathbb{C}^2$ (dimension 2)
- 2 qubits: Hilbert space is $\mathbb{C}^2 \otimes \mathbb{C}^2 = \mathbb{C}^4$ (dimension 4)
- 3 qubits: $\mathbb{C}^2 \otimes \mathbb{C}^2 \otimes \mathbb{C}^2 = \mathbb{C}^8$ (dimension 8)
- n qubits: $(\mathbb{C}^2)^{\otimes n} = \mathbb{C}^{2^n}$ (dimension 2^n)

The state of an n-qubit system is a unit vector in $\mathbb{C}^{2^n}$. The basis states are all 2^n binary strings: for 2 qubits, $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$.

A general 2-qubit state is:

$$|\psi\rangle = \alpha_{00}|00\rangle + \alpha_{01}|01\rangle + \alpha_{10}|10\rangle + \alpha_{11}|11\rangle$$

with $\sum |\alpha_{ij}|^2 = 1$.

What’s new in multi-qubit systems: entanglement

The single-qubit Bloch sphere picture does **not** straightforwardly generalize to multiple qubits. A 2-qubit state lives in $\mathbb{C}^4$, which is not representable as two independent Bloch spheres. The reason: **entanglement**.

A state like $|00\rangle$ is a **product state** — it can be written as $|0\rangle \otimes |0\rangle$. Each qubit has its own independent state.

But the Bell state $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ is **entangled** — it cannot be factored into a product of individual qubit states. The qubits do not have independent descriptions. This is what makes quantum computing powerful (and what makes the state space grow exponentially).

So: the single qubit is the atom, the tensor product is how atoms combine, and entanglement is the qualitatively new phenomenon that emerges when they do.

What does the Hadamard gate actually move?

Claim to evaluate: When I apply a Hadamard gate, the Bloch sphere doesn't move, the state vector doesn't move — rather the basis switches. Apply it again and you're back.

Verdict: This is a valid perspective, but there's an equally valid complementary one. Which you use depends on whether you adopt the "active" or "passive" interpretation of the transformation.

Active picture (state moves, basis stays)

In the standard operational view of quantum computing: the **state moves** and the **basis stays fixed**. The computational basis $\{|0\rangle, |1\rangle\}$ is nailed down (it's what your measurement apparatus reads out). The Hadamard gate rotates the state vector on the Bloch sphere — specifically a 180° rotation about the axis halfway between x and z.

- Start at north pole ($|0\rangle$) $\rightarrow$ H sends it to the +x equator ($|+\rangle$)
- The state physically moved on the sphere; the basis didn't change
- Apply H again $\rightarrow$ the state rotates back to the north pole ($|0\rangle$)

This is how circuit diagrams are conventionally read: gates transform states.

Passive picture (state stays, basis moves)

Your interpretation is the **passive** picture: the state vector is a fixed abstract object in Hilbert space, and the Hadamard changes which basis you use to describe it. Before H, you're expressing the state in the Z-basis. After H, you're expressing the *same* state in the X-basis.

- Before: state described as $\alpha|0\rangle + \beta|1\rangle$ (Z-basis components)
- After H: same state, now described as $\alpha'|+\rangle + \beta'|-\rangle$ $\rightarrow$ but re-expressed in the Z-basis this gives different coefficients
- Apply H again: back to the original description

In this view the Hadamard is a change of coordinate system, not a physical rotation.

Why both are valid

This is the same active/passive duality that appears throughout physics (rotating the object vs. rotating the coordinate axes). Mathematically they are equivalent — a unitary transformation U acting on a state ($U|\psi\rangle$) produces the same measurement predictions as the inverse transformation acting on the measurement basis.

In quantum computing, the **active picture is standard** because: - Circuits are sequences of operations that transform states - Measurement is always in the computational (Z) basis at the end - The basis is physically defined by the measurement apparatus

But the passive picture gives better intuition for *why* $H^2 = I$: switching from Z-basis to X-basis and back is a round trip that returns you to where you started. It also clarifies why H appears at the beginning of many algorithms — you're not "doing something to the qubit" so much as "choosing to look at it in the frequency domain."

Summary

	Active picture	Passive picture
What moves?	State vector rotates	Basis rotates
What's fixed?	Basis (Z-basis)	State vector
H on $\|0\rangle$	State moves to $\|+\rangle$	Description changes from Z to X
$H^2 = I$	Rotation by 180° twice = 360° = identity	Two basis changes = round trip
Standard in QC?	Yes	Less common, but valid for intuition

Pauli gates: rotations on the Bloch sphere

Each Pauli matrix, applied as a gate, performs a **180° rotation** (radians) about its respective axis on the Bloch sphere:

- σ_x (X gate): 180° about the x-axis
- σ_y (Y gate): 180° about the y-axis
- σ_z (Z gate): 180° about the z-axis

This is always 180° regardless of where the state starts on the sphere.

Concrete examples for σ_x (rotation about x-axis by 180°):

Start	End	What happened
$\|0\rangle$ (north pole)	$\|1\rangle$ (south pole)	Flipped through the equator
$\|1\rangle$ (south pole)	$\|0\rangle$ (north pole)	Flipped back
$\|+\rangle$ (+x axis)	$\|+\rangle$ (+x axis)	On the rotation axis — unchanged
$\|-\rangle$ (−x axis)	$\|-\rangle$ (−x axis)	On the rotation axis — unchanged
$\|+i\rangle$ (+y axis)	$\|-i\rangle$ (−y axis)	Flipped across the xz-plane

Why 180°?

Each Pauli matrix satisfies $\sigma^2 = I$ (applying it twice gives the identity). A rotation that returns to the identity after two applications must be 180° — since 180° + 180° = 360° = full rotation = identity.

For rotations by *other* angles, you use the **rotation operators**:

$$R_x(\theta) = e^{-i\theta\sigma_x/2} = \cos\frac{\theta}{2}I - i\sin\frac{\theta}{2}\sigma_x$$

This rotates by angle θ about the x-axis (and similarly for R_y, R_z). The Pauli gate itself is the special case $\theta = \pi$:

$$R_x(\pi) = -i\sigma_x$$

(equal to σ_x up to global phase, hence the same physical operation).

So the Pauli matrices are the "half-turn" gates. For finer rotations you exponentiate them with a continuous parameter.

Pauli matrices: dual role as gate and observable

When a Pauli matrix is applied to a state ($\sigma_x|\psi\rangle$), it is acting as a **linear operator** — a map from the Hilbert space to itself. The term "operator" is correct and precise.

As noted in the Spadework section, the Pauli matrices serve two distinct roles:

- As a **gate**: apply $\sigma_x|\psi\rangle$ to transform the state (e.g. $\sigma_x|0\rangle = |1\rangle$, a bit flip). This uses their unitary property.
- As an **observable**: measure σ_z and get eigenvalue $+1$ (collapse to $|0\rangle$) or -1 (collapse to $|1\rangle$). This uses their Hermitian property.

Same matrix, two interpretations — "where does it send the state?" (gate) vs. "what are its eigenvalues?" (observable).

Q3: Operator / Basis Duality and Reinforcement

This document delves into operators and basis (top half) and then reviews the entire conceptual frame so far (bottom half).

Q2 developed the single-qubit framework: states on the Bloch sphere, gates as rotations, the Hadamard as DFT, and measurement as projection. Here in Q3 we zoom out to see the structural pattern that connects all of this: the operator/basis duality. The same state can be described in different bases, and the Fourier transform (or the Hadamard) is the bridge between conjugate descriptions. This pattern scales from a single qubit to the full continuous theory of position and momentum.

One state, many representations

A quantum state $|\ \ \rangle$ is a single vector in Hilbert space. But we can *express* that vector in different bases, much as a geometric arrow in 3D can be written in Cartesian or polar coordinates without changing the arrow itself.

For a particle:

- **Position basis:** expand $|\ \rangle$ in eigenstates of the position operator → you get $\psi(x)$, the wave function in position space. $|\psi(x)|^2$ gives the probability density of finding the particle at position x.
- **Momentum basis:** expand the same $|\ \rangle$ in eigenstates of the momentum operator → you get $\phi(p)$, the wave function in momentum space. $|\phi(p)|^2$ gives the probability density of the particle having momentum p.

These are not two different states. They are two representations of the same vector, written in two different bases.

The Fourier transform as change of basis

The connection between position and momentum representations is:

$$\phi(p) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} \psi(x)\, e^{-ipx/\hbar}\, dx$$

$$\psi(x) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} \phi(p)\, e^{ipx/\hbar}\, dp$$

The Fourier transform *is* the change-of-basis operation — it maps components in one eigenbasis to components in the conjugate eigenbasis. In finite dimensions this would be a unitary matrix; in infinite dimensions it's an integral transform, but the role is the same.

Conjugate observables and non-commutativity

Position and momentum are **conjugate observables**, linked by the canonical commutation relation:

$$[\hat{x}, \hat{p}] = \hat{x}\hat{p} - \hat{p}\hat{x} = i\hbar$$

This non-commutativity is the origin of the Heisenberg uncertainty principle. If $\psi(x)$ is tightly peaked (position well-known), then its Fourier transform $\phi(p)$ must be spread out (momentum poorly known), and vice versa. This is a mathematical theorem about Fourier transform pairs — the physics is in the assertion that nature uses this structure.

Operators in each basis

Each observable takes a simple form in its own eigenbasis and a complicated form in the conjugate basis:

	In position basis	In momentum basis
Position $\hat{x}$	Multiply by x: $\hat{x}\psi(x) = x\psi(x)$	Derivative: $\hat{x}\phi(p) = i\hbar\frac{\partial\phi}{\partial p}$
Momentum $\hat{p}$	Derivative: $\hat{p}\psi(x) = -i\hbar\frac{\partial\psi}{\partial x}$	Multiply by p: $\hat{p}\phi(p) = p\phi(p)$

This is the **operator / basis duality**: in its own eigenbasis, an operator is diagonal (just multiplication by the eigenvalue). In the conjugate basis, it becomes a differential operator. The Fourier transform swaps which is which.

The qubit analog

For a single qubit the same structure appears in the smallest possible Hilbert space (²):

Continuous system	Qubit system
Position basis {	x }
Momentum basis {	p }
Fourier transform	Hadamard gate
$[\hat{x}, \hat{p}] = i\hbar$	σ_z and σ_x do not commute

The Hadamard gate is the 2-point discrete Fourier transform (up to normalization):

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

It maps Z-eigenstates to X-eigenstates and back:

$$H|0\rangle = |+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$
$$H|1\rangle = |-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$
$$H|+\rangle = |0\rangle, \quad H|-\rangle = |1\rangle$$

Applied twice: $H^2 = I$. Just as the Fourier transform applied twice (with appropriate conventions) returns you to the original, the Hadamard is its own inverse.

> The Hadamard gate is its own inverse (without imposing conjugation) because it is operating on bases of just two elements. The first kernel is the DC mean ($\times$ 1) and the second kernel has a phase of π (or $-\pi$) ($\times$ 1, $\times$ -1) so there is nothing to conjugate. Add a second qubit and now with a four-element basis we no longer call it a Hadamard gate; rather it is a Quantum Fourier Transform (QFT) and we *do* need to conjugate to go from operator to inverse.

Complementarity in the qubit

If a qubit is in state |0 , it has a definite Z-value. Apply H and it becomes |+ — now it has a definite X-value but a completely uncertain Z-value. This is the discrete analog of the uncertainty principle: certainty in one basis implies maximal uncertainty in the conjugate basis.

In quantum algorithms: - **H at the start** creates superposition (explores all computational inputs simultaneously) - **H at the end** causes interference (amplitudes combine constructively for correct answers, destructively for wrong ones)

This is the computational use of the operator / basis duality: you set up a problem in one basis, let quantum dynamics evolve it, and read out the answer in the conjugate basis.

The Pauli matrices and the Bloch sphere

The three Pauli matrices define the three complementary measurement axes for a single qubit:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Any single-qubit state can be written as a point on the **Bloch sphere**, parameterized by two angles and :

$$|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\phi}\sin\frac{\theta}{2}|1\rangle$$

The north pole ($= 0$) is $|0$, the south pole ($=$) is $|1$, the equator contains all equal superpositions (including $|+$ and $|-$). The Pauli matrices generate rotations about the x, y, and z axes of this sphere.

The Hadamard gate corresponds to a 180° rotation about an axis halfway between x and z — which is geometrically why it swaps the Z-basis and X-basis.

Summary of the duality

The central idea: a quantum system has complementary descriptions connected by a unitary transformation (Fourier transform in infinite dimensions, Hadamard in the qubit case). Each description makes one observable simple (diagonal) and the conjugate observable complicated (off-diagonal or differential). Non-commutativity of conjugate operators is not a defect but the structural feature that enables superposition, interference, and ultimately quantum computational advantage.

Connecting back to the ansätze

From Q0's overheard fragments:

- ✓ "There are mathematical (linear) operators corresponding to physical observations" — yes, and they take different forms in different bases.
- ✓ "These operators do not commute" — conjugate operators, connected by Fourier/Hadamard.
- ✓ "This produces a residual quantity tied to the uncertainty principle" — the commutator $[\hat{x}, \hat{p}] = i\hbar$ (or non-commuting Paulis in the qubit case).
- ✓ "There are objects called Pauli spin matrices" — they define measurement axes and generate rotations on the Bloch sphere.

Retrace your steps

This section revisits the single-qubit framework with the objective of reinforcing connections between concepts, addressing gaps that emerged during a knowledge check, and building toward a more complete understanding.

What is a quantum state, really?

A quantum state is a unit vector in a complex Hilbert space. For a single qubit that space is $\mathbb{C}^2$: two complex numbers α and β with $|\alpha|^2 + |\beta|^2 = 1$. Written out:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle = r_\alpha e^{i\eta}|0\rangle + r_\beta e^{i\nu}|1\rangle$$

After factoring out the global phase $e^{i\eta}$ (which changes nothing observable), two physical degrees of freedom remain:

- θ (colatitude): determined by $r_\alpha = \cos(\theta/2)$. This sets the probability balance: $P(|0\rangle) = \cos^2(\theta/2)$, $P(|1\rangle) = \sin^2(\theta/2)$.
- $\phi = \nu - \eta$ (relative phase / azimuth): the phase *difference* between the two coefficients.

These two angles parameterize the Bloch sphere. Every point on its surface is a valid pure state. The poles are the computational basis states; the equator holds all equal-weight superpositions.

Why is the relative phase ϕ physically meaningful?

This was identified as a gap during the knowledge check. The key question: if Z-basis measurement only depends on θ (i.e. $P(|0\rangle) = \cos^2(\theta/2)$, independent of ϕ), then how is ϕ observable at all?

The answer: **measurement in a different basis reveals ϕ.**

The hardware always measures in the Z-basis — it projects onto $|0\rangle$ or $|1\rangle$ and reports a classical bit. But you can *effectively* measure in any basis by applying a gate before the Z-measurement:

To measure in…	Apply this gate first	Then measure Z	Outcome 0 means…	Outcome 1 means…
Z-basis	(nothing)	as-is	state was $\|0\rangle$	state was $\|1\rangle$
X-basis	H	then Z	state was $\|+\rangle$	state was $\|-\rangle$
Y-basis	$S^\dagger H$	then Z	state was $\|+i\rangle$	state was $\|-i\rangle$

Concrete example: Consider the equatorial state $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + e^{i\phi}|1\rangle)$. This has $\theta = \pi/2$ (equator), so Z-measurement always gives 50/50 regardless of ϕ. But if you apply H first:

$$H|\psi\rangle = \frac{1}{2}[(1 + e^{i\phi})|0\rangle + (1 - e^{i\phi})|1\rangle]$$

Now the probabilities are: - $P(|0\rangle) = |1 + e^{i\phi}|^2/4 = \cos^2(\phi/2)$ - $P(|1\rangle) = |1 - e^{i\phi}|^2/4 = \sin^2(\phi/2)$

The relative phase ϕ is now encoded in the measurement statistics. It was always physically real; you just had to ask the right question (measure in the right basis) to see it.

The general principle: "Measuring in basis B" means applying the unitary that maps B-eigenstates to Z-eigenstates, then doing the standard Z-measurement. The hardware never changes; you change what you present to it.

Why complex scalars? The role of interference

The scalar field must be $\mathbb{C}$ (not $\mathbb{R}$) because quantum mechanics requires interference — the ability for probability amplitudes to partially cancel.

With real scalars, two amplitudes can only add constructively ($a+b$) or destructively ($a-b$). With complex scalars, they can combine at any angle: $a+e^{i\phi}b$ gives a continuum of interference patterns parameterized by ϕ. This is what the double-slit experiment demonstrates for single particles.

The formal postulate this encodes: a quantum state is a *complex* superposition of basis states, and the *squared magnitudes* of the complex amplitudes give probabilities (the Born rule). The complex phases between terms produce the interference that distinguishes quantum from classical probability.

Without $\mathbb{C}$: no azimuthal angle on the Bloch sphere, no complementary bases, no Fourier duality, no quantum computational advantage.

The dual role of operators: gates versus observables

The Pauli matrices are both **unitary** and **Hermitian**. This means they serve two distinct roles:

As gates (unitary role): $\sigma_x|0\rangle = |1\rangle$. The operator transforms the state — a 180° rotation about the x-axis. This is deterministic and reversible.

As observables (Hermitian role): "Measure σ_z" means: project the state onto the eigenstates of σ_z (which are $|0\rangle$ and $|1\rangle$) and obtain one of the eigenvalues ($+1$ or -1). This is probabilistic and irreversible.

The eigenvalues of a Hermitian matrix are always real — which is why measurement outcomes are real numbers. The eigenvectors of a Hermitian matrix are orthogonal — which is why measurement outcomes are mutually exclusive.

What does it mean to "measure σ_x"? It means: find the eigenvalues of σ_x ($+1$ and -1) and the eigenstates ($|+\rangle$ and $|-\rangle$). The measurement produces $+1$ (collapsing to $|+\rangle$) or -1 (collapsing to $|-\rangle$), with probabilities determined by the state's projection onto these eigenstates.

Operationally: measuring σ_x is the same as applying H then measuring σ_z. The hardware only implements Z-measurement; everything else is achieved by pre-rotation.

Measurement: what makes it different

Gates are unitary: they preserve the full state (two continuous degrees of freedom). They are deterministic and reversible.

Measurement is a fundamentally different operation: - It is **probabilistic**: you cannot predict the outcome for a single shot. - It is **irreversible**: the state collapses to an eigenstate; the prior superposition is lost. - It is **basis-dependent**: what you learn depends on which observable you measure. - It **extracts classical information**: the output is a bit (0 or 1), not a quantum state.

The reason we repeat circuits many times (shots) is that a single measurement yields one random sample from the probability distribution. The distribution itself — which encodes the answer to the computation — can only be reconstructed from many samples. This is inherently statistical.

To reconstruct the *full* state (both θ and ϕ), you need to measure in *multiple* bases across many shots. This is called **quantum state tomography**: - Z-measurement statistics give θ (the probability balance) - X-measurement statistics give ϕ (the relative phase) - Y-measurement provides a consistency check

No single measurement basis reveals everything. Complementary information lives in complementary bases — this is the uncertainty principle made operational.

The Hadamard as DFT: what does this analogy actually mean?

The 2-point DFT of a vector (x_0, x_1) produces two coefficients: - $X_0 = (x_0 + x_1)/\sqrt{2}$ — the "DC" or average (how much are the two values alike?) - $X_1 = (x_0 - x_1)/\sqrt{2}$ — the "oscillation" (how much do they differ?)

For a qubit state $\alpha|0\rangle + \beta|1\rangle$, applying H produces: - New amplitude for $|0\rangle$: $(\alpha + \beta)/\sqrt{2}$ — "are α and β similar?" - New amplitude for $|1\rangle$: $(\alpha - \beta)/\sqrt{2}$ — "do they differ?"

The Z-basis asks "which position?" The X-basis asks "what frequency?" (i.e. "same or different?"). Transforming between these two questions is exactly what the Fourier transform does, and the Hadamard is the smallest version of it.

In quantum algorithms this is powerful because: - You start in the Z-basis (definite position: $|0\rangle$) - Apply H to enter the X-basis (superposition: $|+\rangle$) - Perform computation (which encodes information in relative phases) - Apply H again to convert phase information back into Z-basis amplitudes - Measure in Z to read out the answer

The information flows from "position" to "frequency" and back, with the computation happening in "frequency space" where parallelism lives.

What the knowledge check revealed

The primary gaps identified were:

1. **Measurement in other bases** — the mechanism by which ϕ becomes observable (apply a rotation, then measure Z). This is now addressed above.
2. **The postulate that demands $\mathbb{C}$** — not a single sentence to memorize but an empirical fact: single particles exhibit self-interference (double slit), which requires continuous phase. Complex superposition is the minimal mathematical structure that supports this.
3. **The DFT analogy** — not nonsensical but initially hard to articulate. The key is that "time domain" = Z-basis = "which slot has the amplitude?" and "frequency domain" = X-basis = "are the amplitudes in phase or out of phase?"
4. **What Pauli matrices measure** — their eigenvalues (± 1) are the measurement outcomes; their eigenvectors define the collapse targets. "Measure σ_z" means "collapse to $|0\rangle$ or $|1\rangle$"; "measure σ_x" means "collapse to $|+\rangle$ or $|-\rangle$."
5. **Why measurement is irreversible** — gates preserve the full state; measurement discards one degree of freedom permanently (within the standard formalism). The connection to information conservation and black holes is genuine but requires more machinery than we have built here.

These are not trivial gaps — they represent the conceptual joints where the formalism bends. Understanding measurement as basis-dependent projection, and understanding that complementary bases reveal complementary information, is the single most important structural insight for what comes next (multi-qubit systems, entanglement, algorithms).

Q4: Entanglement, Controlled Gates, and the Interior of the Bloch Sphere

Q3 established the operator/basis duality and showed how the single-qubit framework scales via tensor products. Q2 introduced entanglement as "the qualitatively new phenomenon" in multi-qubit systems. Here we examine entanglement in detail: what it is mathematically, how controlled gates produce it, and why entangled qubits live inside the Bloch sphere rather than on its surface.

What is entanglement?

The definition

A multi-qubit state is **entangled** if it cannot be written as a tensor product of individual qubit states. That's it — the entire definition is a negative statement about factorizability.

For two qubits, a **product state** (not entangled) has the form:

$$|\psi\rangle = |\phi_A\rangle \otimes |\phi_B\rangle$$

where $|\phi_A\rangle \in \mathbb{C}^2$ and $|\phi_B\rangle \in \mathbb{C}^2$ are independent single-qubit states. Each qubit has a well-defined state of its own.

An **entangled state** is anything that cannot be written this way. The canonical example is the Bell state:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

Try to factor this as $(a|0\rangle + b|1\rangle) \otimes (c|0\rangle + d|1\rangle) = ac|00\rangle + ad|01\rangle + bc|10\rangle + bd|11\rangle$. Matching coefficients: $ac = 1/\sqrt{2}$, $ad = 0$, $bc = 0$, $bd = 1/\sqrt{2}$. From $ad = 0$: either $a = 0$ or $d = 0$. But if $a = 0$ then $ac = 0 \neq 1/\sqrt{2}$. If $d = 0$ then $bd = 0 \neq 1/\sqrt{2}$. Contradiction — the state does not factor.

What entanglement means physically

When two qubits are entangled, neither qubit has a definite state on its own. The system has a definite state (the joint vector in $\mathbb{C}^4$), but the individual qubits do not. This is not ignorance — it's not that each qubit "secretly" has a state that we don't know. The individual states genuinely do not exist. The whole is not the sum of its parts.

Measurement reveals this: in the Bell state $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$, measuring qubit A gives 0 or 1 with equal probability. But once you get the result, qubit B's outcome is determined with certainty — perfectly correlated, regardless of the spatial separation between the qubits. The correlation was established when the entanglement was created, not at the moment of measurement.

How does a CNOT gate produce entanglement?

The CNOT gate

The CNOT (controlled-NOT) gate acts on two qubits: a **control** and a **target**. Its action in the computational basis:

$$\text{CNOT}|00\rangle = |00\rangle, \quad \text{CNOT}|01\rangle = |01\rangle$$
$$\text{CNOT}|10\rangle = |11\rangle, \quad \text{CNOT}|11\rangle = |10\rangle$$

Rule: if the control is $|1\rangle$, flip the target. If the control is $|0\rangle$, do nothing. As a matrix in the $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$ basis:

$$\text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

When CNOT does NOT produce entanglement

If the control qubit is in a computational basis state, CNOT cannot entangle:

- $\text{CNOT}(|0\rangle \otimes |\phi\rangle) = |0\rangle \otimes |\phi\rangle$ — nothing happens. Still a product state.
- $\text{CNOT}(|1\rangle \otimes |\phi\rangle) = |1\rangle \otimes X|\phi\rangle$ — the target gets flipped, but the result is still a product state.

In both cases the output factors. The control has a definite value, so the "conditional" nature of the gate doesn't create any correlation — it either fires or it doesn't, deterministically.

When CNOT DOES produce entanglement

Put the control in a **superposition** first:

$$|+\rangle|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)|0\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$$

Now apply CNOT:

$$\text{CNOT}\left(\frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)\right) = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

The $|00\rangle$ term: control is 0, target untouched $\to |00\rangle$. The $|10\rangle$ term: control is 1, target flipped $\to |11\rangle$.

Result: the Bell state $|\Phi^+\rangle$. We proved above this doesn't factor.

The mechanism: the CNOT created a correlation between the two qubits — "if control is 0 then target is 0; if control is 1 then target is 1." When the control is in superposition, this correlation *is* entanglement. The target's state becomes conditional on the control's state, but the control doesn't have a definite state — so neither does the target. They become a single inseparable unit.

The general principle

A controlled gate produces entanglement if and only if the control qubit is in a superposition (not in a computational basis state). The superposition makes the "which branch?" question indefinite, and the controlled operation correlates the target with that indefinite question. That correlation is entanglement.

This is why the standard Bell-state preparation circuit is: H on qubit A (creates superposition), then CNOT with A as control and B as target (converts superposition into entanglement).

Entanglement and the interior of the Bloch sphere

Pure states live on the surface

For a single isolated qubit, every pure state lives on the **surface** of the Bloch sphere. The surface represents all states with two well-defined degrees of freedom (θ and ϕ). A point on the surface means: the qubit has a definite state — you know everything there is to know about it.

The interior represents mixed states

The **interior** of the Bloch sphere represents **mixed states** — situations where the qubit does not have a definite pure state. The center of the sphere is the **maximally mixed state**: $\rho = \frac{1}{2}I$ — complete ignorance about the qubit.

Points between the center and the surface represent partial knowledge. The distance from the center (the Bloch vector length, $0 \leq r \leq 1$) quantifies how pure the state is: $r = 1$ is pure (surface), $r = 0$ is maximally mixed (center).

How entanglement puts qubits inside the sphere

Here is the connection: when two qubits are entangled, **neither qubit individually has a pure state**. The joint two-qubit system is in a pure state (a definite vector in $\mathbb{C}^4$), but each qubit on its own is described by a **reduced density matrix** — obtained by "tracing out" the other qubit.

For the Bell state $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$:

The reduced density matrix of qubit A (ignoring qubit B) is:

$$\rho_A = \mathrm{Tr}_B(|\Phi^+\rangle\langle\Phi^+|) = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1| = \frac{1}{2}I$$

This is the maximally mixed state — the center of the Bloch sphere. Qubit A has no preferred direction at all. It is equally likely to be found in $|0\rangle$ or $|1\rangle$ (or $|+\rangle$ or $|-\rangle$, or any other basis). The qubit's Bloch vector has collapsed to a point at the origin.

The general rule: the more entangled a qubit is with another system, the further inside the Bloch sphere its individual (reduced) state sits. Maximal entanglement → center of the sphere. No entanglement → surface of the sphere. Partial entanglement → somewhere in between.

Why this happens

A pure state on the surface of the Bloch sphere means: all the qubit's information is locally accessible. You could (in principle) determine its state completely by measuring it alone.

An entangled qubit has "exported" some of its information into correlations with another qubit. That information is nonlocal — it exists only in the joint state, not in either qubit individually. The interior position on the Bloch sphere reflects this loss of local information: the qubit appears random not because we're ignorant, but because its description is genuinely incomplete without reference to its partner.

This is fundamentally different from classical uncertainty. A coin under a cup is either heads or tails — we just don't know which. A maximally entangled qubit is not secretly $|0\rangle$ or secretly $|1\rangle$. It has no individual state at all. The "uncertainty" is ontological, not epistemic.

The density matrix formalism

To describe states inside the Bloch sphere, we need to go beyond kets. The **density matrix** (or density operator) ρ generalizes the state vector:

- **Pure state:** $\rho = |\psi\rangle\langle\psi|$ — a rank-1 projector. Bloch vector has $r = 1$ (on the surface).
- **Mixed state:** $\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$ — a statistical mixture. Bloch vector has $r < 1$ (inside the sphere).

For a single qubit, any density matrix can be written as:

$$\rho = \frac{1}{2}(I + \vec{r} \cdot \vec{\sigma})$$

where $\vec{r} = (r_x, r_y, r_z)$ is the Bloch vector and $\vec{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$. The length $|\vec{r}| \leq 1$, with equality for pure states.

The reduced state of an entangled qubit is a mixed state — not because we've lost information through carelessness, but because the information is *structured* in a way that is inaccessible locally. This is the deepest physical content of entanglement: information that exists but cannot be localized.

Summary of the three-way connection

Condition	Qubit's Bloch vector	State type	Factorizability
Qubit is unentangled	On the surface ($r = 1$)	Pure	Joint state is a product
Qubit is partially entangled	Inside the sphere ($0 < r < 1$)	Mixed	Joint state partially factorizes
Qubit is maximally entangled	At the center ($r = 0$)	Maximally mixed	Joint state is a Bell state

The Bloch sphere's interior is not a mathematical curiosity or a symptom of experimental imprecision. It is where entangled qubits live when viewed individually — and it is the geometric signature that information has become nonlocal.

Entanglement from another Basis

The tensor product $\otimes$ is the operation that constructs multi-qubit systems from single-qubit parts. Understanding it clarifies the boundary between "independent qubits acted on in parallel" and "a genuine multi-qubit operation that can entangle."

Defining the tensor product

For states (vectors): Given qubit A in state $|a\rangle = \begin{pmatrix} a_0 \\ a_1 \end{pmatrix}$ and qubit B in state $|b\rangle = \begin{pmatrix} b_0 \\ b_1 \end{pmatrix}$, their joint state is:

$$|a\rangle \otimes |b\rangle = \begin{pmatrix} a_0 b_0 \\ a_0 b_1 \\ a_1 b_0 \\ a_1 b_1 \end{pmatrix}$$

Every component of the first multiplies every component of the second: a 2-vector tensored with a 2-vector gives a 4-vector. This is the Kronecker product.

Example: $|0\rangle \otimes |1\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} = |01\rangle$

For operators (matrices): If gate A is 2×2 and gate B is 2×2:

$$A \otimes B = \begin{pmatrix} a_{00}B & a_{01}B \\ a_{10}B & a_{11}B \end{pmatrix}$$

Each scalar entry of A becomes a 2×2 block — that scalar times the entire matrix B. Result: a 4×4 matrix that acts on the 4-dimensional joint state space.

The essential property: $(A \otimes B)(|a\rangle \otimes |b\rangle) = (A|a\rangle) \otimes (B|b\rangle)$. The two operators act independently on their respective qubits. No cross-talk.

The joint Hilbert space exists whether or not qubits are entangled

A common confusion: "the 4-dimensional space is for entangled qubits." No — the space $\mathbb{C}^2 \otimes \mathbb{C}^2 = \mathbb{C}^4$ is the state space for *any* two-qubit system, entangled or not. The product state $|00\rangle$ and the Bell state $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ both live in $\mathbb{C}^4$. The space must be large enough to contain all possibilities.

Entanglement is a property of specific *states within* the space — those that don't factor as $|a\rangle \otimes |b\rangle$. The space itself is neutral.

$H \otimes H$ versus the QFT: two very different 4×4 matrices

$H \otimes H$ ("apply H to each qubit independently"):

$$H \otimes H = \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}$$

This factors: it is literally H on qubit A tensored with H on qubit B. It *cannot* create entanglement. Applied to a product state, it produces a product state. It is self-inverse ($(H \otimes H)^2 = I_4$) for the same reason each factor is: nothing to conjugate in either piece.

$\mathbf{QFT}_4$ (the 4-point quantum Fourier transform):

$$\mathrm{QFT}_4 = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -1 \\ 1 & -i & -1 & i \end{pmatrix}$$

This does *not* factor as $A \otimes B$. It is a genuine two-qubit operation. Its kernel entries include i and $-i$, so complex conjugation is nontrivial: $\mathrm{QFT}_4^\dagger \neq \mathrm{QFT}_4$. The inverse transform requires replacing every $e^{-2\pi ijk/4}$ with $e^{+2\pi ijk/4}$ — precisely the sign change in the exponent that distinguishes forward from inverse in the classical DFT.

Why H is self-inverse but QFT is not: The 2-point DFT kernel entries are $\{1, -1\}$ — all real, so conjugation does nothing. At 4 points the kernel acquires genuinely complex entries ($\pm i$), and conjugation is no longer trivial. This is the algebraic content of your observation: "there is nothing to conjugate" at $N = 2$, but there *is* something to conjugate at $N \geq 4$.

Does the QFT have its own operator symbol? Is there a multi-qubit family?

Yes to both.

The QFT is denoted QFT_N (or F_N) where $N = 2^n$ for n qubits. It has a standard circuit decomposition using Hadamard gates and controlled-phase gates (CR_k), requiring $O(n^2)$ gates for n qubits. This is exponentially fewer than the classical FFT's $O(N \log N)$ arithmetic operations — one of the early motivations for quantum computing.

The multi-qubit gate family is large. Here are the main categories:

Category	Examples	Key property
Single-qubit gates applied in parallel	$H \otimes H$, $X \otimes I$	Factor as tensor products; cannot entangle
Controlled gates	CNOT, CZ, Toffoli (CCX)	Conditional operation; *can* entangle
Multi-qubit unitaries	QFT, Grover diffusion operator	Don't factor; operate on the full joint space
SWAP and permutation gates	SWAP, Fredkin (CSWAP)	Rearrange qubits

The controlled gates are the workhorses — CNOT in particular is universal for entanglement when combined with single-qubit rotations. Any unitary on n qubits can be decomposed into single-qubit gates + CNOTs.

Reading states: how to tell entangled from not

Given a state written on the page, how do you determine whether it's entangled?

The factorization test: Can you write $|\psi\rangle_{AB} = |a\rangle_A \otimes |b\rangle_B$? If yes → product state (not entangled). If no → entangled.

Practical examples for two qubits:

State	Entangled?	How to tell
$\|00\rangle$	No	$= \|0\rangle \otimes \|0\rangle$, trivially factors
$\|+\rangle\|0\rangle = \frac{1}{\sqrt{2}}(\|00\rangle + \|10\rangle)$	No	$= \frac{1}{\sqrt{2}}(\|0\rangle + \|1\rangle) \otimes \|0\rangle$, factors
$\frac{1}{\sqrt{2}}(\|00\rangle + \|11\rangle)$	Yes	Try $(a_0\|0\rangle + a_1\|1\rangle)(b_0\|0\rangle + b_1\|1\rangle)$: get $a_0 b_1 = 0$ and $a_1 b_0 = 0$ but $a_0 b_0 \neq 0$ and $a_1 b_1 \neq 0$ — contradiction
$\frac{1}{\sqrt{2}}(\|00\rangle + \|01\rangle)$	No	$= \|0\rangle \otimes \frac{1}{\sqrt{2}}(\|0\rangle + \|1\rangle) = \|0\rangle\|+\rangle$, factors
$\frac{1}{2}(\|00\rangle + \|01\rangle + \|10\rangle + \|11\rangle)$	No	$= \|+\rangle \otimes \|+\rangle$, factors
$\frac{1}{2}(\|00\rangle + \|01\rangle + \|10\rangle - \|11\rangle)$	Yes	Attempt to factor fails (exercise: try it)

The quick heuristic: Write the amplitudes as a 2×2 matrix (rows indexed by first qubit, columns by second qubit). The state is a product state if and only if this matrix has rank 1 — i.e. its determinant is zero.

For $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$: the amplitude matrix is $\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$. Determinant $= \frac{1}{2}(1 \cdot 1 - 0 \cdot 0) = \frac{1}{2} \neq 0 \rightarrow$ rank 2 $\rightarrow$ entangled.

For $\frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$: the amplitude matrix is $\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}$. Determinant $= \frac{1}{2}(0 - 0) = 0 \rightarrow$ rank 1 $\rightarrow$ product state.

This determinant test works for two qubits. For three or more qubits, there is no single number that diagnoses entanglement — you need to check factorizability across every possible bipartition of the system, and there are different *kinds* of entanglement (GHZ vs W states, for example). The subject becomes considerably richer.